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Mullaney

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(54) **REFUSE CONTAINER WITH WEIGHT INDICATOR AND DANGER ALERTING**

(76) Inventor: **Gregory Mullaney**, Cockeysville, MD (US)

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B30B 9/30	(2006.01)
G01G 23/37	(2006.01)
G06Q 10/00	(2012.01)

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CPC ***G01G 19/62*** (2013.01); ***G01G 23/3707*** (2013.01); ***G01G 23/3728*** (2013.01); ***G06Q 10/30*** (2013.01); *B30B 9/3007* (2013.01)

(58) **Field of Classification Search**
CPC G01G 19/62; G01G 23/3707; G01G 23/3728; B30B 9/3007; G06Q 10/30
USPC 177/238–245, 25.13, 177, 45
See application file for complete search history.

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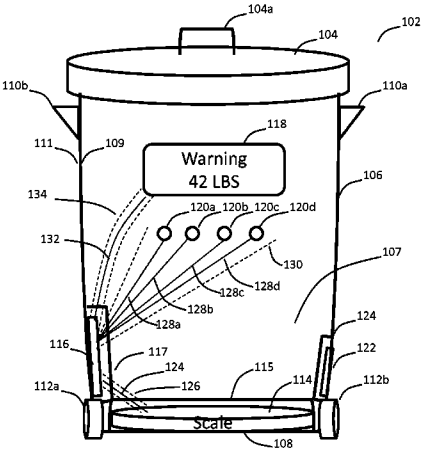
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Primary Examiner — Randy W Gibson
(74) *Attorney, Agent, or Firm* — David C. Isaacson; Fitzgerald & Isaacson, LLP

(57) **ABSTRACT**

A refuse container provides an indication of the weight of the refuse stored therein and provides a danger alert when the weight of the refuse nears or exceeds a predetermined maximum. A display can display a value based on the weight of the refuse held in the refuse container. Further, one or more light sources can provide an indication of the weight of the refuse held in the refuse container. For example, the number of light source lit can provide an indication of the weight of the refuse held in the refuse container. A scale in the refuse container measures the weight of the refuse held therein. A controller uses the weight of the refuse held in the refuse container to determine the value to display on the display and whether to activate the light sources.

18 Claims, 5 Drawing Sheets



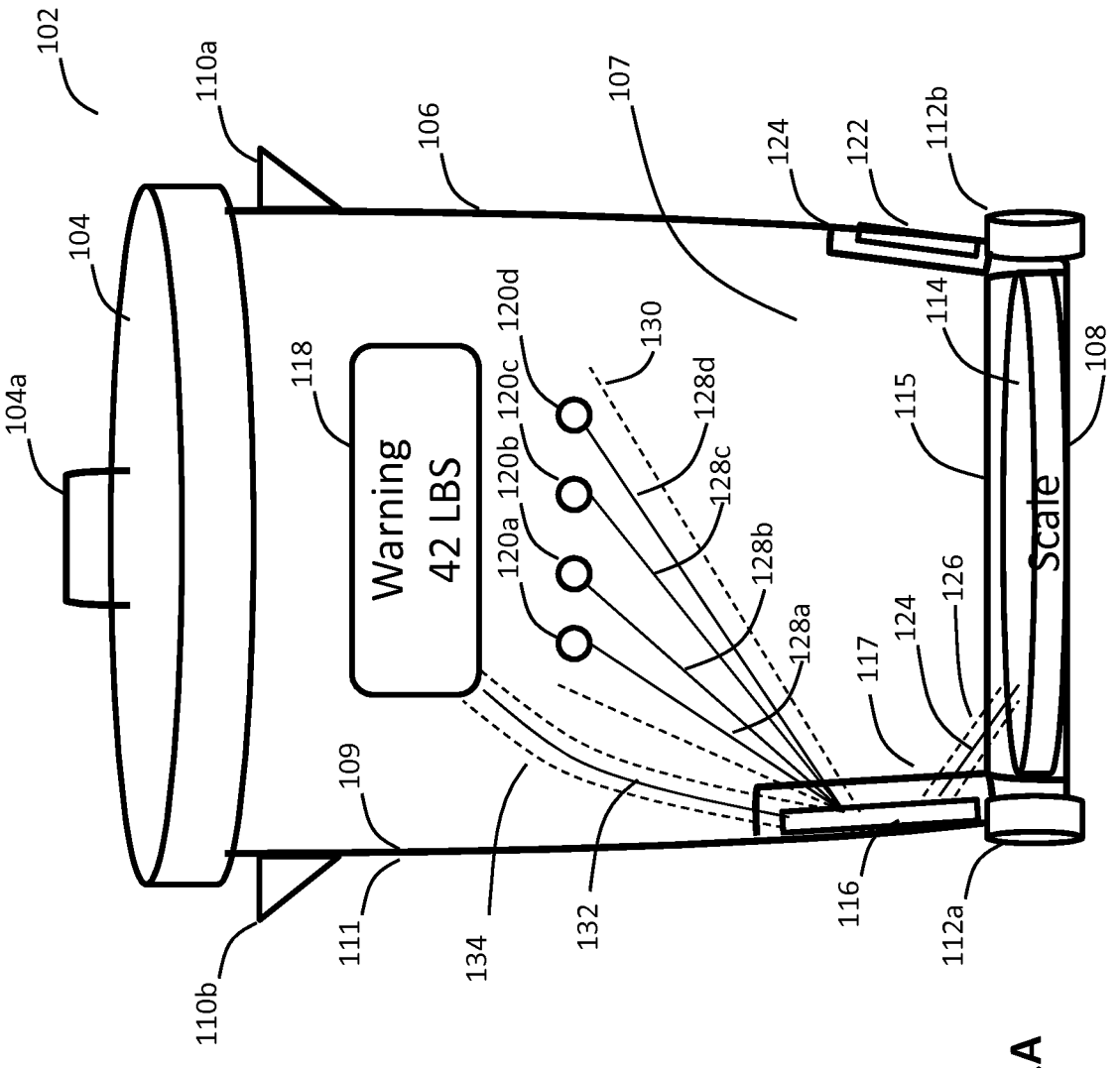

102
110a
106
107
124
122
112b
104
114
108
104a
118
Warning
42 LBS
120d
120c
120b
120a
130
128d
128c
128b
128a
115
Scale
126
124
117
109
111
134
132
116
112a
110b

Figure 1A

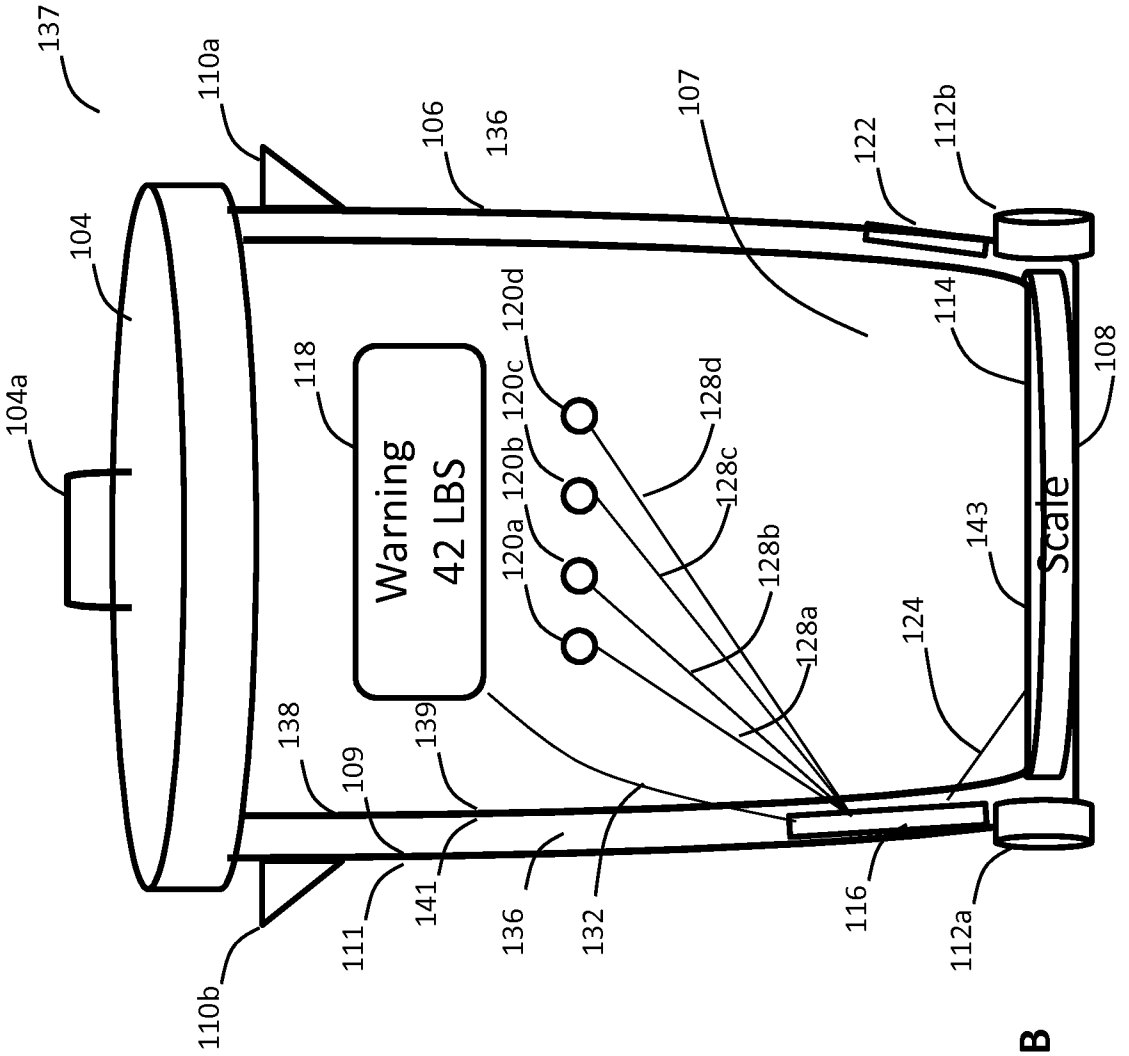

137
110a
106
136
107
122
112b
104
104a
118
Warning
42 LBS
120d
120c
120b
120a
128d
128c
128b
128a
114
108
143
Scale
124
138
109
139
111
141
136
132
116
112a
110b

Figure 1B

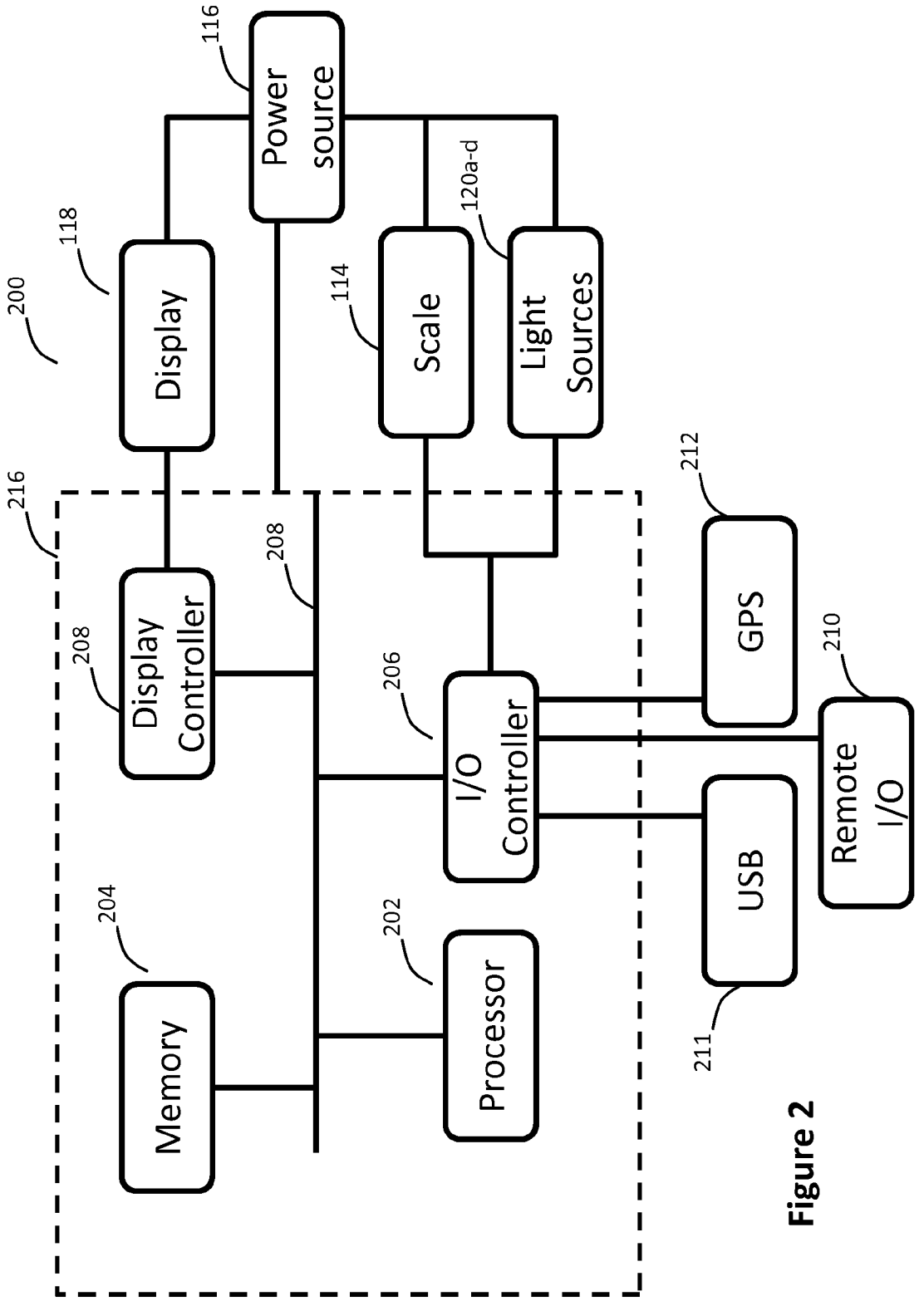

200
116
Power source
118
Display
114
Scale
120a-d
Light Sources
216
208
Display Controller
208
206
I/O Controller
212
GPS
210
Remote I/O
204
Memory
202
Processor
USB
211

Figure 2

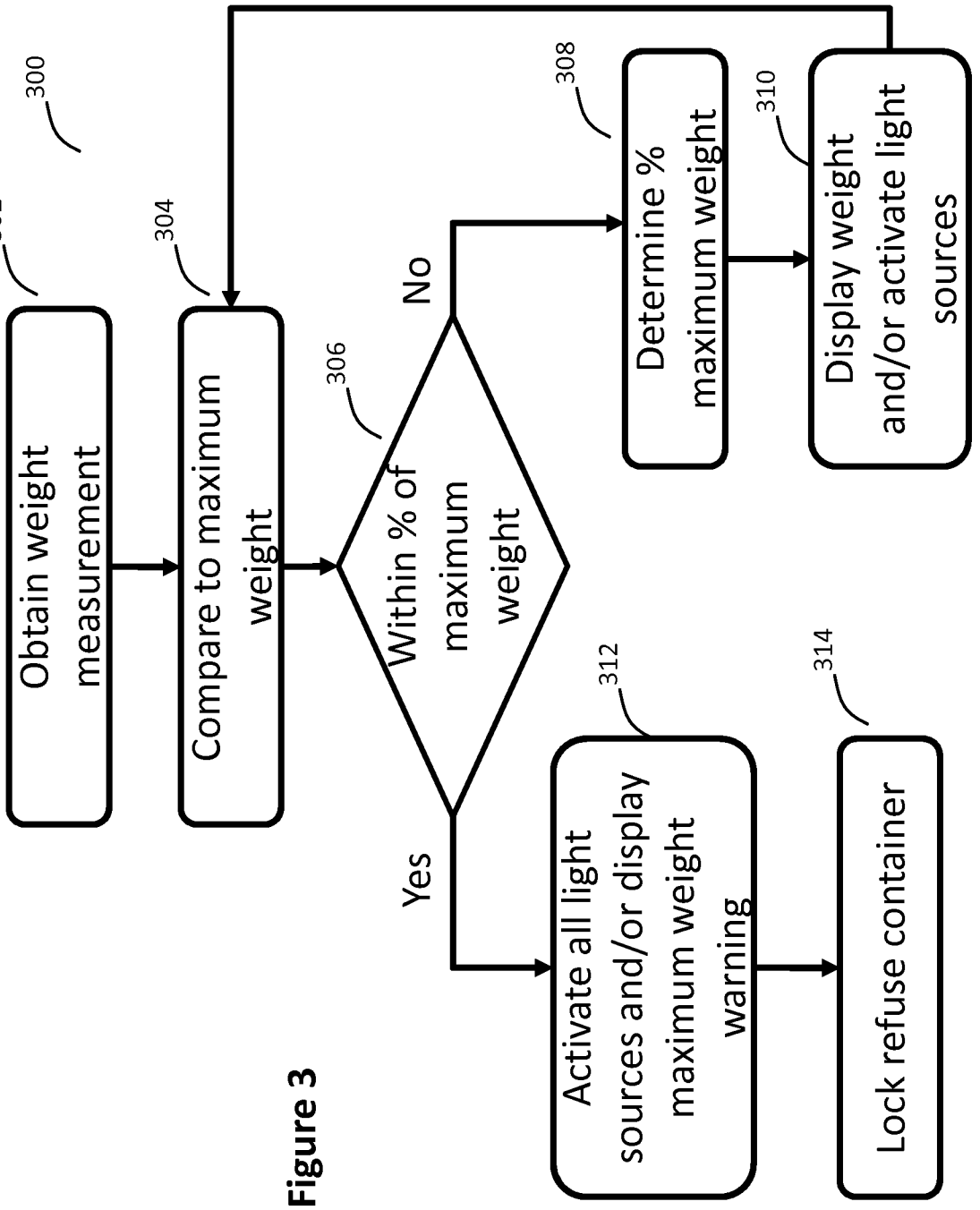

300
302
Obtain weight measurement
304
Compare to maximum weight
306
Within % of maximum weight
No
Yes
308
Determine % maximum weight
310
Display weight and/or activate light sources
312
Activate all light sources and/or display maximum weight warning
314
Lock refuse container

Figure 3

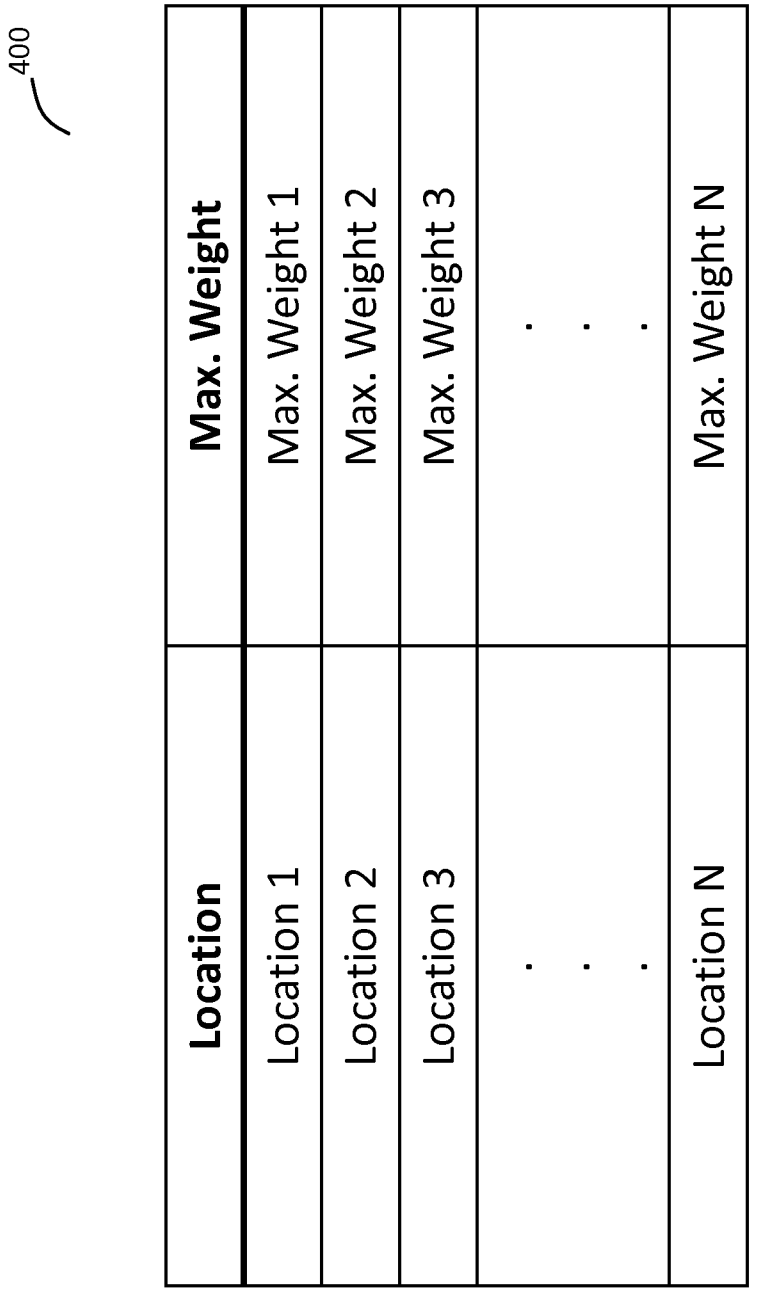

400
Location
Max. Weight
Location 1
Max. Weight 1
Location 2
Max. Weight 2
Location 3
Max. Weight 3
Location N
Max. Weight N

Figure 4

REFUSE CONTAINER WITH WEIGHT INDICATOR AND DANGER ALERTING

BACKGROUND

1. Field of the Invention

Embodiments of the present invention relate generally to healthy and safety. More specifically, embodiments of the present invention relate to a refuse receptacle that provides an alert to prevent dangerous weight conditions.

2. Background of the Invention

Those involved on the frontlines of refuse collection have a physically demanding job. Moving and lifting heavily loaded garbage cans or other refuse containers is hard work. Typically, there is no way for a refuse collector to know how heavy a particular garbage can is prior to moving or picking it up. How full a particular refuse container is may provide little information as to its weight. For example, a refuse container full of paper products may weigh very little, whereas a refuse container only partially full of more dense refuse such as metal or liquid product may be very heavy. Thus, it is virtually impossible for a refuse collector to know what effort to apply to move or lift a particular refuse container. As a result, refuse collectors are prone to injury, particularly if they are surprised by the weight of a refuse container they attempt to move or lift.

Such injuries affect refuse collectors and their employers alike. Refuse collectors suffer pain, recoupment time, lost wages, and potentially shortened careers. Employers suffer lost workers, increased insurance costs, and inefficiencies by having to replace workers or ask current workers to work overtime. Of course, requesting that current workers work overtime makes those workers more prone to fatigue and injury, thereby spiraling the cycle

Some jurisdictions responsible for waste management provide regulations concerning refuse. For example, such jurisdictions may provide that refuse containers not exceed 50 or 60 pounds per container. However, there is little chance of anyone filling a refuse container to have any idea of the weight of refuse already in the container or how much additional weight can be added to the container without violating a jurisdictional regulation.

BRIEF SUMMARY OF THE INVENTION

In an embodiment, a refuse container provides an indication of the weight contained therein as well as an alert to warn a refuse collector when a maximum weight is nearing or has been exceeded.

In an embodiment, the refuse container comprises a body in which refuse is held, a scale to measure the weight of the refuse held in the body, and a controller to cause an alert to be provided based on the weight of the refuse. For example, the alert can be provided when the weight of the refuse exceeds a percentage of a predetermined maximum weight or percentage thereof. In addition to or in lieu of the alert, an indication of the weight of the refuse being held in the refuse container can be provided. In an embodiment, a display is provided to allow display the weight of the refuse in the refuse container. In an embodiment, light sources are activated in accordance with the weight of the refuse stored in the container. In an embodiment, the predetermined maximum weight is determined based on a location of the refuse container.

Another embodiment is a method for providing an alert in a refuse container. The method includes measuring a weight of refuse held in the refuse container as a measured weight using a scale, comparing the measured weight to a predetermined maximum weight, and providing an alert based on the comparison. In addition to or in lieu of the alert, an indication of the weight of the refuse being held in the refuse container can be provided. In an embodiment, the method includes displaying the weight of the refuse in the refuse container. In an embodiment, the method includes activating light sources accordance with the weight of the refuse stored in the container. In an embodiment, the method include determining the predetermined maximum weight on a location of the refuse container.

In an embodiment, a table containing one or more entries, each entry having a location and a corresponding maximum weight is stored in a memory. In an embodiment, to determine the predetermined maximum weight, a location of the refuse container is determined. The determined location is compared to the locations in the table. If there is a match the maximum weight stored in the entry with the location that matched the determined location of the refuse container is used as the predetermined maximum. If there is no match, a user can be prompted for a maximum weight to use or a default maximum weight can be used.

Additional features and embodiments of the present invention will be evident in view of the following detailed description of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**A illustrates an exemplary refuse container having an alert mechanism according to an embodiment of the present invention.

FIG. **1**B illustrates an exemplary refuse container having an alert mechanism according to an alternate embodiment of the present invention.

FIG. **2** is a schematic diagram of a system for providing alerting for a refuse container according to an embodiment of the present invention.

FIG. **3** is flow chart for controlling a refuse container display and light source(s) according an embodiment of the present invention.

FIG. **4** is an exemplary table for storing locations and corresponding maximum weight values according to an embodiment of the present invention.

DETAILED DESCRIPTION

FIG. **1**A illustrates an exemplary refuse container **102** having an alert mechanism according to an embodiment of the present invention. In an embodiment, refuse container **102** includes a lid **104**. In an embodiment, lid **104** includes a handle **104***a*. Lid **104** can be detachable from body **106**. Lid **104** can also be attached to body **106** via a hinge to allow the lid to be opened and closed.

Refuse container **102** also includes a body **106**. Body **106** can be formed with one or more walls. The wall(s) forming body **106** have an inner surface **109** and an outer surface **111**. Body **106** can be formed of any suitable material, including, without limitation rubber, metal, plastic, composite material, or combinations thereof. Body **106** encloses an inner volume **107**. Refuse container **106** also includes a floor **108**. In an embodiment, refuse container **102** includes one or more wheels **112***a* and **112***b* to facilitate movement of refuse container **102**. In an embodiment, refuse container **102** includes one or more handles, for example, handles **110***a* and **110***b*. Handles **110***a* and **110***b* can be used for lifting or moving refuse container **102**.

According to an embodiment, refuse container **102** includes a scale **114**. Scale **114** can be any device that can be

used to measure a weight of refuse placed on top of it. In an embodiment, scale **114** is calibrated when ready to be deployed such that scale **114** registers zero weight with no refuse in the container. Scale **114** can also be calibrated to take into consideration other weights, such as garbage bag weight for garbage bags that will hold refuse that is placed in refuse container **102**. Scale **114** can be placed on top of floor **108**, partially embedded in floor **108**, or fully embedded in floor **108**. Further, in an embodiment, floor **108** can comprises a bottom surface and a top surface, and scale **108** can be placed between the bottom and top surfaces of floor **108** such that the top surface of floor **108** covers scale **114**. In an embodiment, scale **114** is protected by a protective covering **115**. Protective covering **115** should allow accurate weight measurement by scale **114**.

Refuse container **102** includes a controller **116** that is used to provide control function for refuse container **102** as described herein. In an embodiment, controller **116** is protected by a protective covering **117**.

In an embodiment, refuse container **102** includes a display **118**. In an embodiment, display **118** provides a digital readout. In an embodiment, the digital readout represents the weight of refuse in refuse container **102** as measured by scale **114**. In another embodiment, the digital readout represents the amount of weight that can still be added to refuse container **102** without exceeding a preset limit. In an embodiment, display **118** is configured to provide a warning message when the weight of refuse in refuse container **102** exceeds a preconfigured maximum weight, or percentage thereof.

In an embodiment, refuse container **102** includes one or more light sources **120***a-d*. Four light sources **120***a-d* are illustrated in the exemplary refuse container **102** for purposes of description. However, any number of light sources can be used for a particular implementation. Light sources **120***a-d* can be lamps, light emitting diodes (LEDs), or any other light source. Embodiments can be configured to have a display, such as display **118**, one or more light sources, such as light sources **120***a-d*, or both a display **118** and one or more light sources.

In an embodiment having a plurality of light sources **120***a-d*, each light source **120***a-d* is the same color. In an alternate embodiment having a plurality of light sources **120***a-d*, each light source **120***a-d* is a different color. In yet another embodiment having a plurality of light sources **120***a-d*, subsets of light sources **120***a-d* are the same color, while different subsets of light sources **120***a-d* are different colors.

In an embodiment, with only one light source, for example light source **120**s, activation of the light source indicates that a maximum weight limit has been reached and no more refuse can be placed in refuse container **102** without exceeding the weight limit. In an alternate embodiment, activation of the light source indicates that a maximum weight limit has not been reached and additional refuse can be added to refuse container **102**.

In an embodiment, controller **116** receives weight measurements from scale **114** and uses those weight measurements to control display **118** and light sources **120***a-d* to indicate weight of the refuse in refuse container **102** and to provide danger alerting. In an embodiment, communications between controller **116** and scale **114** occur over a wire **124**. In an embodiment, wire **124** is located wholly between inner surface **109** and outer surface **111**. For protection, a protective conduit **126** can be provided to allow wire **124** to be run from controller **116** to scale **114** through protective conduit **126**. In an embodiment, protective conduit **126** is molded in or attached to body **106**. Protective conduit **126** can be made from the same material(s) as body **106** or other suitable material. In an embodiment, protective conduit **126** is located wholly between inner surface **109** and outer surface **111**. In an alternate embodiment, communication between controller **116** and scale **114** is wireless. In such an embodiment, protective conduit **126** may not be required.

In an embodiment, communications between controller **116** and light sources **120***a-d* occur over wires **128***a-d* respectively. For protection, a protective conduit **130** can be provided to allow wires **128***a-d* to be run from controller **116** to light sources **120***a-d* through protective conduit **130**. In an embodiment, wires **128***a-d* are located wholly between inner surface **109** and outer surface **111**. In an embodiment, there is a conduit **130** for each wire **128***a-d*. In an embodiment, protective conduit **130** is molded in or attached to body **106**. Protective conduit **130** can be made from the same material(s) as body **106** or other suitable material. In an embodiment, protective conduit **130** is located wholly between inner surface **109** and outer surface **111**. In an alternate embodiment, communication between controller **116** and light sources **120***a-d* is wireless. In such an embodiment, protective conduit **130** may not be required.

In an embodiment, communications between controller **116** and display **118** occur over a wire **132** respectively. In an embodiment, wire **132** is located wholly between inner surface **109** and outer surface **111**. For protection, a protective conduit **134** can be provided to allow wire **132** to be run from controller **116** to display **118** through protective conduit **134**. In an embodiment, there is a conduit **134** for wire **132**. In an embodiment, protective conduit **134** is molded in or attached to body **106**. Protective conduit **134** can be made from the same material(s) as body **106** or other suitable material. In an embodiment, protective conduit **134** is located wholly between inner surface **109** and outer surface **111**. In an alternate embodiment, communication between controller **116** and display **118** is wireless. In such an embodiment, protective conduit **134** may not be required.

Refuse container **102** also includes a power source **122**. Power source **122** provides power to the components of refuse container **102** that may need it. For example, in an embodiment, power source **122** is used to provide power to components such as display **118**, light sources **120***a-d*, scale **114**, and controller **116**. Other components may require power in a particular implementation. Power source **122**, can be any power source that can be configured to provide the required power, including batteries and AC power sources. In an embodiment, power source **122** is protected by a protective covering **124**.

In use, refuse is placed within inner volume **107** and on floor **108** or directly on scale **114** such that the weight of the refuse is measured by scale **114**. Measurement of refuse weight by scale **114** causes a read out to be displayed on display **118**. In addition, one or more light sources **120***a-d* may be activated depending on the weight of refuse in refuse container **102** measured by scale **114**. As described above, in one embodiment, the read out and lights sources are related to the weight of refuse that is in refuse container **102**. Alternately, the read out and light sources indicate that additional refuse can be added to refuse container **102**.

FIG. **1**B illustrates an exemplary refuse container **137** having an alert mechanism according to another embodiment of the present invention. Like numbered items in FIG. **1**A are referenced similarly in FIG. **1**B. Refuse container **137** includes an inner shell **138**. Inner shell **138** can be a separate unit from body **106** or a part of body **106**. Further, inner shell **138** and body **106** can be formed from a unitary material. Inner shell **138** can be the same material as body **106** or any other suitable material. Inner shell **138** can be formed with

one or more walls. Like the wall(s) of body **106**, the wall(s) of inner shell **138** has an inner surface **139** and an outer surface **141**.

In an embodiment, inner shell **138** includes a floor **143**. Floor **143** must be configured to allow accurate weight measurements by scale **114**. In another embodiment, inner shell **138** does not have a floor. In such embodiment, the wall(s) of inner shell **138** can be affixed to scale **114** such that scale **114** effectively forms a floor for inner shell **138**. In such a case, a protective covering, such as protective covering **115** can be used to protect scale **114**.

As can be seen in FIG. **1**B, inner surface **109** of body **106** and outer surface **141** of inner shell **138** form a space **136**. In an embodiment, space **136** is an empty space. In an embodiment, space **136** is filled with a material. In an embodiment, controller **116**, power source **122**, and wires **124**, **128***a-d*, and **132** are wholly contained within space **136**. In this manner the materials of body **106** and inner shell **138** provide protection to these components. Scale **114** may fit partially or wholly within space **136**. For example, scale **114** is partially in space **136** when it acts as an effective floor for inner shell **138** as described above. Light sources **120***a-d* and display **118** are partially within space **136**. These components are protected by the materials of body **106** and inner shell **138** at least to the extent they are included in space **136**. In an embodiment, one or more conduits are included in space **136** through which to run wires such as wires **120***a-d*, **124**, and **132**.

Body **106** of refuse containers **102** and **137** can be any suitable shape. Such shapes include, for example and without limitation, circular, square, rectangular, oval, diamond, and square or rectangular with rounded corners. Further, inner shell **138** can be, but need not be, the same shape as body **106**.

FIG. **2** is a schematic diagram of an exemplary system **200** to provide alerting for a refuse container according to an embodiment of the present invention. System **200** includes a controller **216**. Exemplary controller **116** of FIGS. **1**A and **1**B can be a controller **216** as illustrated in FIG. **2**. It should be understood that an appropriate controller for use in embodiments is not limited to the structure illustrated in FIG. **2**. As such in an embodiment, a controller for use in embodiments may have more or fewer components than illustrated in FIG. **2**.

As illustrated in FIG. **2**, in an embodiment, controller **216** includes a processor **202**. Processor **202** can be a microprocessor, microcontroller, or any other device that can be configured to control alerting in a refuse container as described herein. Controller **216** also includes a memory **204**. Memory **204** stores program code and other information required to allow processor **202** to perform the functions described herein. Memory **204** can also include external memory, such as CD-ROM, memory stick, hard disk, and other external memories. Memory **204** can be in one or more memory devices, and thus can be any combination of internal and/or external memory. Further memory **204** or portions thereof can be coupled to bus **208** directly and/or indirectly. Not all of memory **204** need be part of controller **216** and embodiments of controller **216** may rely solely on external memory.

Controller **216** includes an input/output (I/O) controller **206**. I/O controller **206** inputs measurements from scale **108** and outputs signals to light sources **120***a-d*. Controller **216** also includes a display controller **208**. Display controller **208** provides signals to control display **118**.

In an embodiment, controller **216** includes a bus **208**. Bus **208** provides for communication between various components of controller **216**, such as, for example, processor **202**, memory **204**, I/O controller **206**, and display controller **208**. Controller **216** can be configured as one or more devices depending upon application, and may have more or fewer components than shown in FIG. **2**.

In operation, processor **202** compares weight values measures by scale **114** to a maximum weight and causes panel display **118** and/or light sources **120***a-d* to be activated accordingly. To determine the maximum weight, a configuration phase is performed. The configuration phase can be performed by the manufacturer and/or by an end user. In one embodiment, a configuration phase is entered into by causing processor **202** to enter into the configuration phase. In an embodiment, this is accomplished using a remote controlling device that provides a configuration signal that is detected by remote I/O detector **210**. In an embodiment, when remote I/O detector **210** detects a remote configuration signal, it causes an interrupt to processor **202** that causes processor **202** to be placed in a configuration state or phase. In an embodiment, the remote signal can be supplied by a remote controller, a mobile phone have an application executing thereon to perform a configuration as described herein, or any other device to provide a remote control signal for configuration.

In an embodiment, in the configuration state or phase **202**, processor **202** causes panel display **118** to display a menu from which one or more choices for set up can be chosen. In one embodiment, for example, display **118** displays, "(1) City/State" then "(2) Zip code", and then "(3) Work Place." Different or additional selections may be available in different embodiments depending on implementation. For example, county, township, area code, or other location information may be used.

If display **118** does not have a sufficient number of lines to display the menu in total, the choices can be displayed in succession and with a hold time. For example, if display **118** has only one line, the three exemplary choices provided above could be displayed one at a time in succession and with a 2-3 second hold time.

If a user chooses 1, processor **202** causes panel display **118** to prompt the user to input a city and state via the remote controller. If the user chooses 2, processor **202** causes panel display **118** to prompt the user to input a zip code. If the user chooses 3, processor **202** causes panel display **118** to prompt the user to input a work place, for example, an employer name or an address.

In an embodiment, in which a mobile phone has an application executing thereon for configuring a refuse container according to an embodiment, the application executing on the mobile phone causes a menu with appropriate choices to be displayed to the user. They user then selects the desired choice, and is prompted to input appropriate information corresponding to the user's choice, for example, a city and state, a zip code, or an employer's name or address. The information is then transmitted for use by processor **202** through remote I/O device **210**.

In another embodiment, refuse container **102** includes a port, such as a USB port that can be coupled to a computing device, such as a laptop, desktop, tablet, mobile telephone, or other computing device that can be coupled to processor **202** through the port. In such a case, an application executing on the computing device provides a menu with appropriate choices to be displayed to the user. The user then selects the desired choice, and is prompted to input appropriate information corresponding to the user's choice, for example, a city and state, a zip code, or an employer's name or address. The information is then transmitted for use by processor **202** through USB port **211**.

In another embodiment, refuse container **102** includes a global positioning system (GPS) device **212**. Other determining location devices can be used as well. GPS device **212**

determines location of refuse container **102**. In an embodiment with GPS device **212**, during the configuration phase, processor **102** obtains location information from GPS device **212**. Such location information can include, but is not limited to, latitude and longitude information.

The user's input in response to the prompt is compared against pre-stored data to obtain a maximum weight. For example, in an embodiment, a table of city/state pairs and corresponding weights can be stored in memory **204**. Similarly, a table of zip codes and corresponding maximum weights can be stored in memory **204**. Similarly, a table of company names and/or addresses and corresponding weights can be stored in memory **204**. In general, a table contained any location data for a particular application and corresponding maximum weight information can be stored.

The table of location data and corresponding maximum weights can be configured by a manufacturer or provided during the configuration phase. The location data can be any information that can be used to provide a location. In either case, the table can be stored in memory **204**. FIG. **4** is an exemplary table **400** for associating location data with maximum weight data. As shown in FIG. **4**, in an embodiment, table **400** includes a set of entries, each entry including location information, for example, locations **1** through N, and corresponding maximum weight data, in this case, maximum weights **1** through N, respectively.

In an embodiment, a number of tables, such as table **400**, are included. Each of the included tables has a different location data type with corresponding maximum weight entries. For example, one table may have zip codes; another may have cities, another may have towns; another may have latitude and longitude, and so forth.

In operation according to an embodiment, an appropriate user input in response to a prompt is compared against the location data of the entries in the table of location data and corresponding maximum weight values to find a match. For example, if the user enters a zip code, processor **202** compares the use entered zip code to zip codes in a table of zip codes and corresponding maximum weight values. In an embodiment with GPS device **212**, the location information obtained from GPS device **212** is compared to the entries in the table. In an embodiment, the GPS information is converted to another location type, for example, zip code, city, town, or street address.

The maximum weight corresponding to the matching entry is retrieved. The retrieved maximum weight is then stored in a maximum weight location in memory. The obtained maximum weight can be stored in a permanent fashion, and updated only upon initiation of the configuration phase described above.

If there is no matching entry, the user is again prompted to enter the information for the user's prompt selection. In addition, the user can also select enter a user-defined maximum weight or choose to use a default maximum weight. In an embodiment, the user can select the default maximum weight as an option on the menu. In an alternative embodiment, the default maximum weight is used if the user fails some number of times to make a matching entry, for example, after 3 tries.

In an embodiment having a GPS device **212**, when there is no matching entry, the user need not be prompted to reenter location information. Instead, the user selects whether to enter a maximum weight or use a default. In an embodiment, the user can select the default maximum weight as an option on the menu. In an alternative embodiment, the default maximum weight is used if there is no matching entry to location information provided by GPS device **212**.

Embodiments having a GPS device **212** can include an operational mode in which they operate as described above with respect to embodiments not having a GPS device **212**. That is, in such a mode, the embodiment will operate as if there were no GPS device **212**. This can be useful if GPS device **212** is not operating correctly for some reason, such as weather or technical issues that interfere with the ability to acquire or track the GPS satellite signal.

FIG. **3** is a flow chart **300** for controlling display **118** and light source(s) **120***a-d* to provide an alert concerning the weight of refuse in refuse container **102** according an embodiment. In step **302** a current weight measurement is obtained from scale **114**. The current weight measurement can be stored in memory **204**. In an embodiment, the current weight measurement is stored with a current date and time stamp. In an embodiment, one or more prior weight measurements, along with their corresponding date and time stamps, are stored in memory **204** to provide a history of weight measurements provided by scale **114**.

Processor **202** then continues in step **304** in which the obtained current weight measurement is compared against the maximum weight measurement stored in memory **204**. If the maximum weight is not exceeded, in step **308**, the percentage of maximum weight of the weight of the refuse in refuse container **102** as measured by scale **114** is determined. In step **310**, the weight of refuse in container **102** and/or the percentage of maximum weight is displayed on display **118**. In addition, as described below one or more light sources **120** are activated in accordance with the percentage of maximum weight of the refuse in refuse contained **102**. In an embodiment, a higher number of light sources **120** activated indicates a greater weight of refuse in refuse container **102**. For example, if the percentage of maximum weight of the refuse in refuse container **102** is determined to be 75%, then three light sources are activated in a four-light source embodiment.

If there is not an exact percentage of light sources to activate then the number of light sources to activate can be rounded to the nearest higher or lower representable percentage. For example, if the percentage of maximum weight of refuse in refuse container **102** was 85%, then in an embodiment, three light sources in a four-light source embodiment are activated as 75% is the nearest lower representable percentage. In an alternative embodiment, if the percentage of maximum weight of refuse in refuse container **102** was 85%, then in an embodiment, four light sources in a four-light source embodiment are activated as 100% is the nearest higher representable percentage.

In an alternate embodiment, the difference between the maximum weight and the weight of the refuse in refuse container **102** is displayed on display **118**. Displaying the difference in this manner indicates the weight of refuse that can still be placed in refuse container **102** without exceeding the maximum weight. In addition, one or more light source **102** is illuminated in accordance with the weight difference. In the alternate embodiment, a higher number of lights sources **120** that are illuminated indicates that a lesser weight of refuse can be added.

Other methods for indicating the weight of refuse in refuse container **102** and activating representative light sources are possible, including, for example, and without limitation, weight percentage remaining, where more activated light sources indicate higher remaining refuse capacity. This may prove useful in saving battery life in cases where refuse container **102** is more often full than not.

If in step **306**, processor **202** determines that the maximum weight has been exceeded, processor **202** continues in step **312** where processor **202** causes panel display **312** to display

the weight contained in refuse container **102** and causes all light sources **120***a-d* to illuminate. In an embodiment, at least one of light sources **120***a-d* can be caused to flash when the weight of the refuse in the refuse container exceeds the maximum weight. In an alternate embodiment, processor **102** causes panel display **118** to display the maximum weight to indicate that the refuse in container **102** has reached or exceeded the maximum weight.

In an alternate embodiment, where the difference is used, panel display will display the difference. In an embodiment, the displayed difference can be 0 or negative number to indicate by how much the refuse in container **102** exceeds the maximum weight. In another embodiment, panel display **118** displays 0 to indicate that the refuse in container **102** has reached or exceeded the maximum weight.

In an embodiment, processor **202** continues in step **314** where it locks lid **104** to prevent additional refuse from being added to container **102**. The locking mechanism can be any mechanism that can be used to lock lid **104**, including mechanical, electro-mechanical, and magnetic locking mechanisms. In an embodiment, refuse container **102** includes a sensor to determine when refuse container **102** is partially or completely inverted. Processor **202** causes the locking mechanism to be released when refuse container **102** is partially or completely inverted to allow emptying of refuse container **102**. In another embodiment the locking mechanism can be released with the push of a button or by manually releasing the locking mechanism. Step **314** is optional and may not be included in some embodiments.

While comparing against the maximum weight measurement in step **304** is acceptable in some embodiments, there is a possibility that using the actual maximum weight will result in exceeding the maximum weight. As a result, in an embodiment, a percentage of maximum weight is used for comparison in step **306**. For example, if the maximum weight in a particular application is 50 pounds, the percentage may be 90%. In that case, the comparison maximum weight used in step **304** would be 45 pounds rather than 50 pounds. Using a percentage of maximum weight in this manner provides a cushion so that exceeding the maximum weight is less likely.

In an embodiment, the percentage of maximum is configured by calculating the percentage maximum weight and storing that calculated value in a percentage maximum weight location in memory. Thereafter, processor **202** uses the percentage maximum weight rather than maximum weight, except when displaying a maximum weight value. In an alternate embodiment, the maximum weight value stored in the maximum weight location in memory **204** is replaced with the percentage maximum weight value.

In an embodiment, the percentage of maximum weight can be changed by the user during the configuration phase. In an embodiment, a default percentage can be stored. For example, the default percentage may be 100% or 95% depending upon application.

As described above, a history of weight measurements and corresponding date and time stamps can be stored in memory **204**. In an embodiment, these historical weight measurements can be accessed remotely by a service provider and analyzed. For example, the analysis may be used to determine users that consistently overload refuse container **102**. In such a case, a notice can be sent to the user to advise the user to avoid overloading refuse container **102** in the future, or that service fees will be increased. In addition, in an embodiment, the notice can be provided on display **118**. Further, the analysis can be used to determine the amount of refuse a particular user is generating so that appropriate fees can be assessed.

Remote access can be by transmission of the historical data over Wi-Fi hotspot or any other technique to transmit the historical data to the service provider, including without limitation cable, cellular telephone network, and satellite transmission. In an embodiment, historical weight data can be transmitted to a service provider according to any schedule, whether periodic or non-periodic. In an embodiment, the historical data is deleted after transmission of the historical data to the service provider.

Weight measurements of the refuse in refuse container **102** can be made according to any schedule, and can be periodic and non-periodic. Examples of periodic weight measurements include, without limitation, measuring the weight once a week, once a day, once an hour, once a minute, etc. Non-periodic weight measurements include, without limitation, measurements according to a non-periodic schedule stored in memory **204**. Another example of a non-periodic weight measurement is to configure controller **216** can be configured to accept remote control commands to cause a weight measurement to be performed. In this manner, a service provider can cause a weight measurement at any desired time. In another exemplary non-periodic weight measurement paradigm, a sensor on lid **104** can be configured to trigger a weight measurement when the lid is removed, thereby indicating either additional refuse is being added to refuse container **102** or refuse container **102** is about to be emptied.

The alert provided by a refuse container as described herein is not limited to visual alerts. Audible alerts can be provided in embodiments. For example, a beep or other audible alert can be provided when the weight of the refuse in the container nears or reaches a predetermined limit.

The foregoing disclosure of the preferred embodiments of the present invention has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many variations and modifications of the embodiments described herein will be apparent to one of ordinary skill in the art in light of the above disclosure. The scope of the invention is to be defined only by the claims appended hereto, and by their equivalents.

Further, in describing representative embodiments of the present invention, the specification may have presented the method and/or process of the present invention as a particular sequence of steps. However, to the extent that the method or process does not rely on the particular order of steps set forth herein, the method or process should not be limited to the particular sequence of steps described. As one of ordinary skill in the art would appreciate, other sequences of steps may be possible. Therefore, the particular order of the steps set forth in the specification should not be construed as limitations on the claims. In addition, the claims directed to the method and/or process of the present invention should not be limited to the performance of their steps in the order written, and one skilled in the art can readily appreciate that the sequences may be varied and still remain within the spirit and scope of the present invention.

What is claimed is:

1. A refuse container, comprising:

a body in which refuse is held;

a lid;

a scale to measure a weight of the refuse being held in the body of the refuse container;

a memory in which is stored a history of the weights that are measured by the scale;

a controller to cause an alert to be provided based on the weight of the refuse, wherein the refuse container is configured to transmit the stored weight history to a service provider; and

a locking mechanism to lock the lid to prevent further refuse from being placed in the refuse container when the weight of refuse in the refuse container reaches a predetermined weight.

2. The refuse container of claim **1**, further comprising a display to display a value determined in accordance with the weight of the refuse being held in the body of the refuse container.

3. The refuse container of claim **2**, wherein the value displayed on the display on the display is the weight of the refuse in the container.

4. The refuse container of claim **2**, wherein the value displayed on the display is based on the weight of refuse that can be placed in the body without exceeding a predetermined maximum weight.

5. The refuse container of claim **2**, wherein the value displayed on the display is based on the difference between a predetermined maximum weight and the weight of the refuse in the body.

6. The refuse container of claim **2**, wherein the alert comprises a warning message displayed on the display when the weight of the refuse held in the body of the refuse container nears a predetermined maximum weight.

7. The refuse container of claim **1**, further comprising one or more light sources that are activated in accordance with the weight held in the body of the refuse container.

8. The refuse container of claim **7**, wherein the number of the one or more light sources that are activated is determined in accordance with the proportion of the weight of refuse in the body of the refuse container to a predetermined maximum weight.

9. The refuse container of claim **7**, wherein the alert comprises flashing at least one of the one or more light sources.

10. The refuse container of claim **1**, wherein the scale is calibrated.

11. The refuse container of claim **1**, wherein the alert is based on how close the weight is to a predetermined maximum weight.

12. The refuse container of claim **1**, wherein the memory comprises:

a table that has at least one entry that contains a location and a corresponding maximum weight; wherein the controller determines a location of the refuse container and uses the determined location as a look up into the table find a table entry having a location matching the determined location, and uses the corresponding maximum weight stored in the table entry with the matching location as the criterion for providing an alert.

13. A method for providing an alert in a refuse container, comprising:

measuring a weight of refuse held in the refuse container as a measured weight using a scale;

storing a history of the measured weights in a memory;

transmitting the history of stored weights to a service provider;

comparing the measured weight to a predetermined maximum weight;

providing an alert based on the comparison; and

locking a lid of the refuse container to prevent further refuse from being placed in the refuse container when the weight of refuse in the refuse container reaches a predetermined weight.

14. The method of claim **13**, further comprising providing the alert when the measured weight exceeds the predetermined maximum weight.

15. The method of claim **13**, further comprising providing the alert when the measured weight is within a percentage of the maximum weight.

16. The method of claim **13**, further comprising displaying a value on a display in accordance with the measured weight.

17. The method of claim **13**, further comprising activating one or more light sources in accordance with the measured weight.

18. A refuse container, comprising:

a body in which refuse is held;

a scale to measure a weight of the refuse being held in the body of the refuse container;

a memory in which is stored a history of the weights that are measured by the scale,

wherein the memory comprises a table that has at least one entry that contains a location and a corresponding maximum weight; wherein the controller determines a location of the refuse container and uses the determined location as a look up into the table find a table entry having a location matching the determined location, and uses the corresponding maximum weight stored in the table entry with the matching location as the criterion for providing an alert; and

a controller to cause an alert to be provided based on the weight of the refuse, wherein the refuse container is configured to transmit the stored weight history to a service provider.

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